

Pigeon Hole Principle

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§1 Problems

§1.1 Level 1

1. Prove that among 370 persons, there will always be two persons having the same birthday.
2. Prove that among any 12 integers there are two which leaves the same remainder when divided by 11.
3. There are n persons present in a room. Prove that among them there are two persons having the same number of acquaintances in the room.
4. Show that any convex polyhedron has two faces with the same number of edges.
5. Let (x_i, y_i) , $1 \leq i \leq 5$, be a set of five distinct points with integer coordinates in the $x - y$ plane. Show that the mid-point of the line joining at least one pair of these points has integer coordinates.
6. Given 51 distinct positive integers strictly less than 100, prove that some two of them sum to 99.

§1.2 Level 2

1. Twenty pairwise distinct positive integers are all less than 70. Prove that amongst the pairwise (positive) differences there are four equal numbers.
2. Prove that from 52 positive integers we can select two such that their sum or difference is divisible by 100.
3. Each of 10 segments is longer than 1 cm and shorter than 55 cm. Prove that you can select three sides of a triangle among the segments.
4. Let S be a set of 25 points such that in any three element subset of S , there are at least two points with distance less than 1. Prove that there exists a 13 element subset of S which can be covered by a disk of radius 1.

5. Prove that two of six points placed into a 3×4 rectangle will have distance not greater than $\sqrt{5}$.
6. There are four points inside a 8×8 square. Prove that two of these points are at $\sqrt{65}$ apart.
7. Prove that in any convex $2n$ -gon, there is a diagonal not parallel to any side.
8. Let $n \geq 3$ be an odd number. Show that there is a number in the set

$$\{2^1 - 1, 2^2 - 1, \dots, 2^n - 1\}$$

which is divisible by n .

9. [**Bezout's Theorem**] Let $a, b \in \mathbb{N}$ be coprime. Then prove that there exist $x, y \in \mathbb{N}$ such that $ax - by = 1$.
10. Thirty-three rooks are placed on a 8×8 chessboard. Prove that you can choose 5 of them which are not attacking each other.
11. Prove that among any seven real numbers $y_1, y_2, \dots, y_7$, there exist two such that

$$0 \leq \frac{y_i - y_j}{1 + y_i y_j} \leq \frac{1}{\sqrt{3}}$$

12. Given m integers $a_1, a_2, \dots, a_m$, show that there exist integers k, s with $0 \leq k < s \leq m$ such that $a_{k+1} + a_{k+2} + \dots + a_s$ is divisible by m .

§1.3 Level 3

1. A chess master who has 11 weeks to prepare for a tournament decides to play at least one game every day, but not more than 12 games during a week. Show that there exists a succession of days during which the chess master will have played exactly 21 games.
2. Given a set of 10 distinct two digit numbers. Prove that you can always choose two disjoint non-empty subset such that the sum of elements is same in both the subsets.
3. Prove that two of 70 distinct positive integers less than or equal to 200 have differences of 4, 5, or 9.
4. An algorithm is a finite sequence of moves performed on a Rubik's cube. Prove that, starting from a solved cube, if you perform an algorithm finitely many times, you will reach the solved state again.

5. Let $A = \{a_1, a_2, \dots, a_{2k+1}\}$, where $k \geq 1$, be a set of $2k+1$ positive integers. Show that for any permutation $b_1, b_2, \dots, b_{2k+1}$ of A , the product

$$\prod_{i=1}^{2k+1} (b_i - a_i)$$

is always even.

6. Consider a prime number p . Every second, you can add either the digit 1 or the digit 3 to the right end of the number (after the unit digit), such that the new number is also a prime. Can this process continue indefinitely?
7. If 11 integers are selected from the set $S = \{1, 2, \dots, 100\}$, prove that there exist at least two integers $x, y \in S$ such that

$$0 < |\sqrt{x} - \sqrt{y}| < 1.$$

§1.4 Level 4

1. Prove that one of $n+1$ numbers from $\{1, 2, \dots, 2n\}$ is divisible by another.
2. Let n be a positive integer and A be a set of $8n+1$ positive integers, all coprime to 6 and less than $30n$. Prove that there exist two distinct numbers $a, b \in A$ such that a divides b .
3. [**Dirichlet's Theorem**] Let α be an irrational number and N be a positive integer. Show that there exist m, n with $m \leq N$ so that $|m\alpha - n| < 1/N$.
4. Each of nine straight lines divide a square into 2 quadrilaterals with the ratio of their area equal to $r > 0$. prove that at least three of these lines are concurrent.
5. Numbers 1 to 81 are written on the squares of a 9×9 board. Prove that there exists two neighbours (i.e. they share an edge) which differ by at least 6.
6. Consider the Fibonacci series $1, 1, 2, 3, 5, 8, \dots$. Show that for any $n \geq 1$, there is a Fibonacci number ending with n zeroes.
7. [**Thue's Lemma**] Let $n > 1$ be an integer and a be an integer coprime to n . Prove that there exist integers x, y with $0 < |x|, |y| < \sqrt{n}$ so that

$$x \equiv ay \pmod{n}$$

8. Prove that no 7 integers, not exceeding 24, can have sums of all subsets different.

§1.5 Level 5

1. Let k be a positive integer and $n = 2^{k-1}$. Prove that from $2n - 1$ positive integers, one can select n integers such that their sum is divisible by n .
2. Prove that in a room with $(m - 1)n + 1$ persons, there are m mutual strangers or there is a person who is acquainted with n persons.
(The above problem can also be solved using *Turan's Lemma* (Graph Theory section)).
3. The length of each side of convex quadrilateral is less than 24. Let P be any point inside $ABCD$. Prove that there exist a vertex, say A , such that $|PA| < 17$.
4. All the points in the plane are colored using three colors. Prove that, there exists a triangle with vertices, having the same color, such that, either it is isosceles or its angles are in geometric progression.
5. In a circle of radius 16, there are 650 points. Prove that there exists a ring of inner radius 2 and outer radius 3, which contains not less than 10 of the given points.
6. [**Simplified version of Ramsey's Theorem**] Prove that in a party of 6 persons there are always three of them who know each other, or three of them such that no two of them know each other.
7. In a group of 17 scientists each scientist sends letters to the others. In their letters only three topics are involved and each couple of scientists makes reference to only one topic. Show that there exists a group of three scientists who send each other letters on the same topic.
8. Suppose 2016 points of the circumference of a circle are colored red and the remaining points are colored blue. Given any natural number $n \geq 3$, prove that there is a regular n -sided polygon all of whose vertices are blue.
(Can be solved without PHP also)

§1.6 Level 6

1. The sides of a equilateral triangle are bicolored. Do there exist, on its perimeter, three monochromatic vertices of a right angled triangle?
2. The n positive integers $a_1 \leq a_2 \leq \dots \leq a_n \leq 2n$ are such that the least common multiple of any two of them is greater than $2n$. Prove that $a_1 > \lfloor 2n/3 \rfloor$.
3. The plane is colored in two colors. Prove that there exists a rectangle with monochromatic vertices. Prove the same result if plane is colored with $k > 2$ colors.
4. Let $a_1, a_2, \dots, a_{100}$ and $b_1, b_2, \dots, b_{100}$ are two permutations of the integers from 1 to 100. Prove that among the products $a_1b_1, a_2b_2, \dots, a_{100}b_{100}$, there are two with the same remainder upon division by 100.

5. Let $\pi(n)$ denote the number of primes at most n . Show that if we have a subset of $\{1, 2, \dots, n\}$ of size $\pi(n) + 1$, some number divides the product of all others.
6. Given a set M of 1985 distinct positive integers, none of which has a prime divisor greater than 23, show that there exists at least one subset of four distinct elements in M whose product is the fourth power of an integer.
7. Consider a set of 16 points arranged in a 4×4 square grid formation. Prove that if any 7 of these points are colored blue, then there exists an isosceles right-angled triangle whose vertices are all blue.
8. From a set of 11 square integers, show that one can choose 6 numbers $a^2, b^2, c^2, d^2, e^2, f^2$, such that

$$a^2 + b^2 + c^2 \equiv d^2 + e^2 + f^2 \pmod{12}$$

9. In a meeting there are representatives of n countries ($n \geq 2$) sitting at a round table. It is known that for any two representatives of the same country their neighbors to their right cannot belong to the same country. Find the largest possible number of representatives in the meeting.
10. In the congress, three disjoint committees of 100 congressmen each are formed. Every pair of congressmen may know each other or not. Show that there are two congressmen from different committees such that in the third committee there are 17 congressmen that know both of them or there are 17 congressmen that know neither of them.
11. Let X be a set of 11 integers. Prove that one can find a non-empty subset $\{a_1, a_2, \dots, a_k\}$ of X such that 3 divides k and 9 divides the sum $\sum_{i=1}^k 4^i a_i$.

§1.7 Level 7 (Personal favourites!)

1. [**Erdos-Szekeres theorem**] For $n \geq (p-1)(q-1) + 1$, prove that every sequence of n integers contain a monotonically increasing subsequence of length p or a monotonically decreasing subsequence of length q .
2. [**Blichfeldt's Lemma**] Let M be a set in the plane with area greater than 1. Show that M contains two distinct points (x_1, y_1) and (x_2, y_2) such that $x_2 - x_1$ and $y_2 - y_1$ are integers.
3. There is a circle of radius 1. A man with irrational step length α (measured along the circumference) walks around the circle. The circle has a ditch of width $\epsilon > 0$. Prove that, sooner or later, he will step into the ditch no matter how small ϵ will be.

4. Prove that a real number α is irrational if and only if there exist two sequences of integers $p_1, p_2, \dots$ and $q_1, q_2, \dots$ such that

$$\lim_{n \rightarrow \infty} q_n \alpha - p_n = 0$$

and $q_n \alpha - p_n \neq 0$ for all n .

5. [**Confinement Lemma**] Given complex numbers $z_1, z_2, \dots, z_n$ in the open unit disk, show that there exist numbers $\epsilon_l = \pm 1$, $l = 1, 2, \dots, n$ such that

$$\left| \sum_{i=1}^m \epsilon_i z_i \right| \leq \sqrt{3},$$

for each $m = 1, 2, \dots, n$.

6. Let $z_1, \dots, z_n$ be complex numbers such that $|z_1| + \dots + |z_n| = 1$. Show that there exists a set $S \subseteq \{z_1, \dots, z_n\}$ with

$$\left| \sum_{z \in S} z \right| \geq \frac{1}{4}.$$